

Air University



Digital Logic Design (Lab)

Year 2025

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Experiment No. 05

Title: Basic Adder Circuit

Objectives:

- To understand the basic principles of a fundamental adder circuit.
- To design and implement a basic adder circuit using logic gates.
- To verify the circuit's performance through a truth table and a Proteus simulation.
- To construct the basic adder circuit on a Logic Trainer Kit.

Equipment and Components:

- Proteus Software (for simulation)
- Logic Trainer Kit (for hands-on execution)
- Logic Gate ICs (7408 - AND, 7486 - XOR)
- Breadboard
- Power Supply (5V)
- Connecting Wires
- Multimeter (optional)

Introduction:

A basic adder circuit is a cornerstone of digital electronics and serves as a building block for more complex arithmetic functions. This circuit is designed to add two binary digits, producing a sum and a carry output. It plays a critical role in arithmetic logic units (ALUs) that perform binary calculations. The construction of this adder circuit employs logic gates, particularly XOR and AND gates, which facilitate efficient binary addition. Understanding the design and function of this circuit is essential for progressing toward more advanced digital systems. This experiment aims to design, construct, and validate a basic adder circuit using both simulation tools like Proteus and practical implementation on a Logic Trainer Kit.

Procedure:

1. Design and Simulation:

Design: Use one XOR gate and one AND gate for the circuit.

Implementation: Connect the inputs A and B to the XOR gate for the sum (S) and to the AND gate for the carry (C).

Verification: Utilize Proteus to simulate the circuit and confirm the results using a truth table.

2. Implementation on Logic Trainer Kit:

Setup:

- Position the ICs (7486 for XOR and 7408 for AND) on the Logic Trainer Kit.
- Connect the VCC and ground to the appropriate pins on the kit.

Circuit Connections:

- Attach inputs A and B to the input switches available on the kit.
- Link the outputs to the output LEDs on the trainer.

Testing:

- Experiment with various input combinations as outlined in the truth table.
- Observe the status of the LEDs to ensure each input combination is correctly indicated.
- Optionally, disconnect the outputs from the LEDs to take voltage readings with a multimeter.

Logic Expression and Truth Table:

Logic Expressions:

- Sum (S) = $\bar{A} \cdot B + A \cdot \bar{B} = A \oplus B$
- Carry (C) = $A \cdot B$

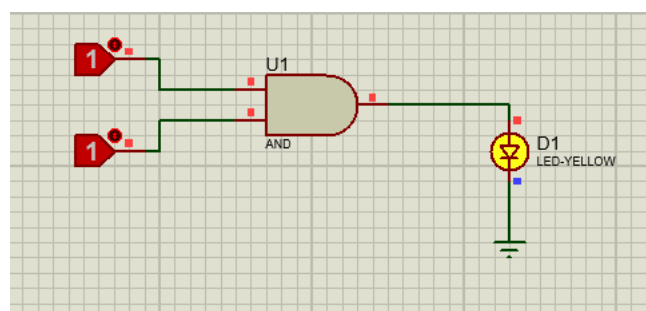
Truth Table:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

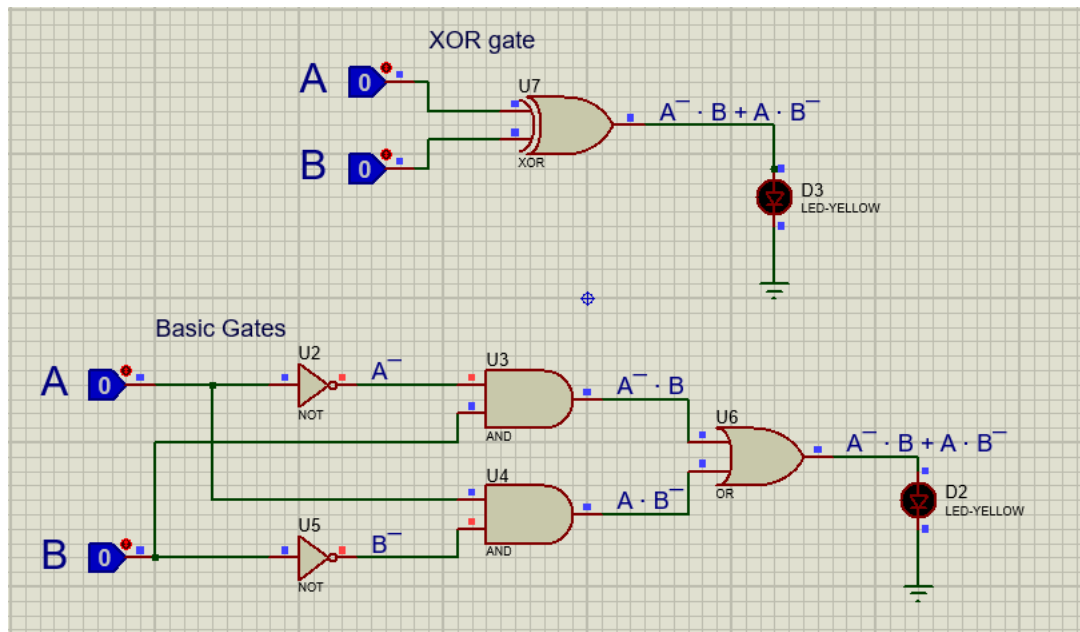
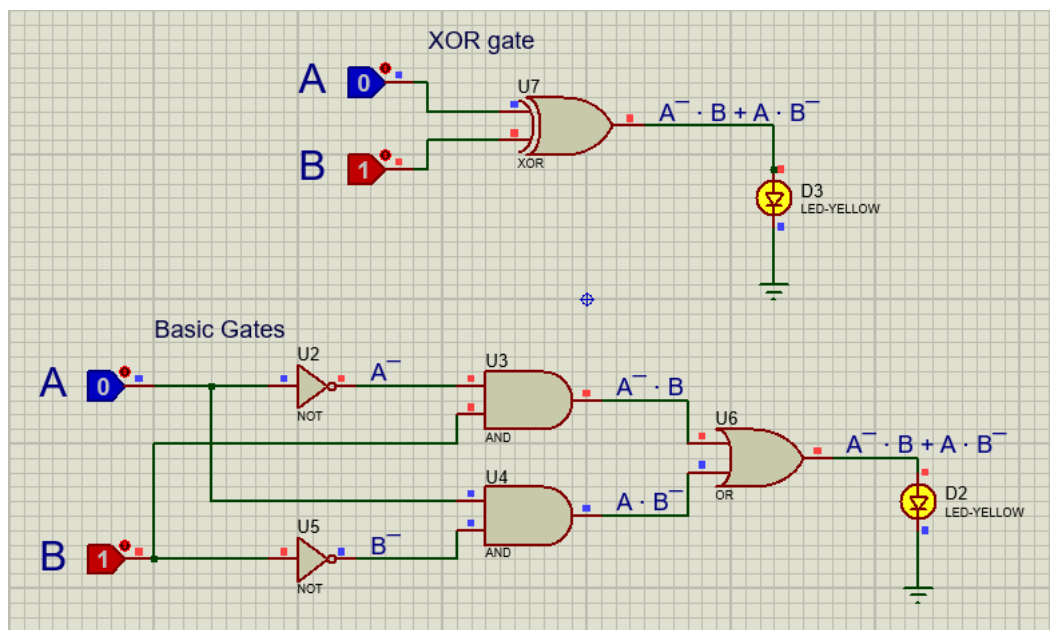
Simulated Results:

i. Verifying: Carry (C) = $A \cdot B$

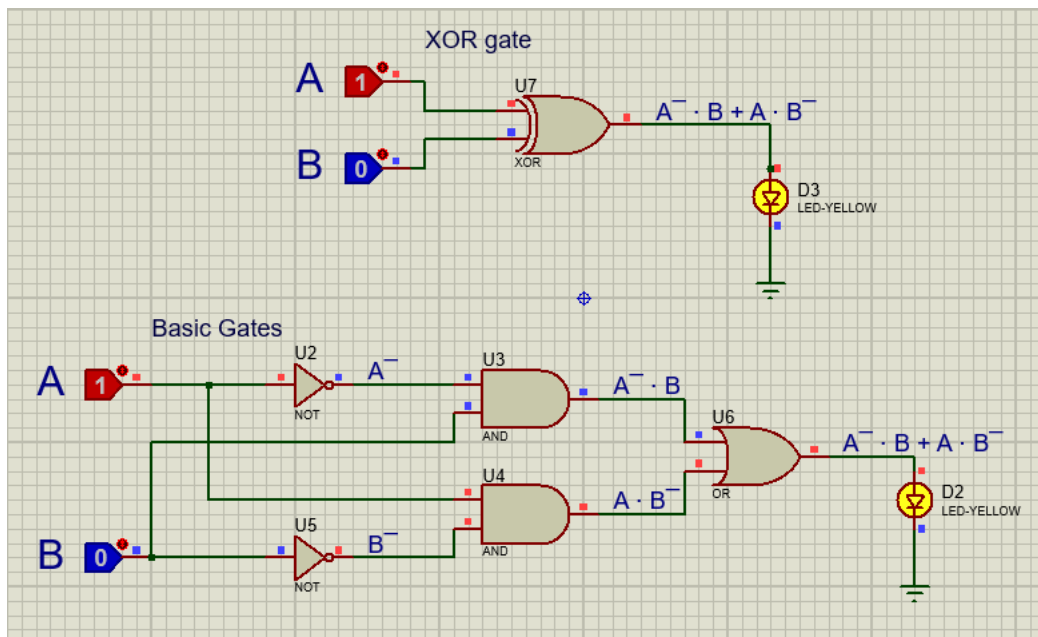
Input: A=1, B=1, → Output=1



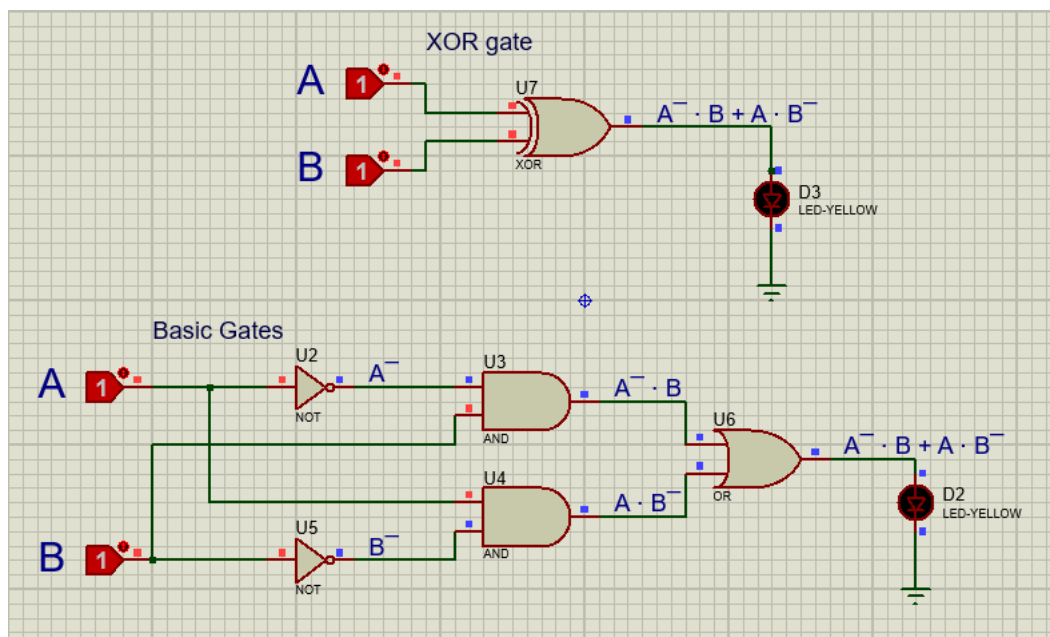
For all other input combinations the output will be Low (0).

ii. Verifying: Sum (S) = $\bar{A} \cdot B + A \cdot \bar{B} = A \oplus B$ **Input: A=0, B=0, $\rightarrow$ Output=0****Input: A=0, B=1, $\rightarrow$ Output=1**

Input: A=1, B=0, → Output=1



Input: A=1, B=1, → Output=0



Conclusions:

This experiment successfully demonstrated the design and implementation of a basic adder circuit using XOR and AND gates. The truth table and Proteus simulation confirmed the circuit's correctness. The principles underlying binary addition were illustrated, highlighting the importance of this circuit in the realm of digital electronics. However, one limitation of this basic setup is its lack of a carry-in bit, which restricts its use to straightforward binary addition. Future experiments could explore more complex adder designs to enhance performance in digital systems.